M. Sc. Physics (Semester-II) 2026
Quantum Mechanics II – Tutorial 3

1. Consider the addition of two angular momenta $j_1 = 1$ amd $j_2 = \frac{1}{2}$.

   (a) Determine all possible values of the total angular momentum j.

   (b) List all the allowed values of quantum number m for each j. Here m is quantum number associated with the eigenvalue equation for operator $J_z = J_{z1} + J_{z2}$ as $J_z|jm\rangle = m\hbar|jm\rangle$.

   (c) Find the dimensionality of combined Hilbert space: $\mathcal{H}_{j_1} \otimes \mathcal{H}_{j_2}$

   (d) Express all the six states $|jm\rangle$ formed by $j_1 = 1$ and $j_2 = 1/2$ as linear combinations of $|j_1 j_2; m_1 m_2\rangle$. Explicitly write the corresponding CG coefficients.

2. Consider the state $|j_1\, j_2; j\, m\rangle$, which is a common eigenstate of the angular momentum operators J_1^2, J_2^2, J^2, and J_z, where $\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$.

   (a)Show that this state is also an eigenstate of the operator $\mathbf{J}_1 \cdot \mathbf{J}_2$ and determine its corresponding eigenvalues.

   (b)Perform the same analysis for the operators $\mathbf{J} \cdot \mathbf{J}_1 \quad$ and $\quad \mathbf{J} \cdot \mathbf{J}_2$.

3.